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Jena, 09.06.2025

Letter of Recommendation: Xiaoqing Pan (潘晓清)

Leibniz Institute for Natural Product
Research and Infection Biology
Hans Knöll Institute

Dear Hiring Committee,

Directorate
Prof. Dr. Axel Brakhage
Elke Jäcksch

I write to you today to recommend Ms. Xiaoqing Pan for the position of Postdoctoral Researcher in your research group. As the head of an independent junior research group and the supervisor of Ms. Pan's doctoral thesis work over the past four years, I believe I am suited to assess her qualifications as a researcher.

Incorporated Society: Jena
Amtsgericht Jena VR 230298

Ms. Pan is currently in the final stages of her doctoral research in my group at the Leibniz Institute for Natural Product Research and Infection Biology: Hans Knöll Institute in Jena, Germany with her degree to be awarded ultimately through her association with the Friedrich Schiller University Jena. In my laboratory, we study RNA regulation in human fungal pathogens with the aim of revealing novel targets for antifungal therapy and improved diagnosis. Much of our work is performed at the bench, but as RNA research often requires analysis of large datasets, we also spend significant effort applying computational tools to better understand the data we generate.

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During her tenure in the lab, Ms. Pan has been involved in numerous aspects of our studies and made important contributions in each case. She came to the laboratory on a German Academic Exchange Service (DAAD) Fellowship after completing her master's studies in her home country of China. There, her studies focused on circRNA identification in plants, with an introduction to both wet and dry-lab work. Her initial efforts in my group were in the wet lab, but she quickly realized that her true passion was in more computational aspects of the project. She therefore dedicated herself towards improving her bioinformatic skills and was able to gain the necessary computational skills both by leveraging existing knowledge in our group and by establishing her own support network at our research institute.

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In her thesis work, Ms. Pan has used her wet-lab and computational skills to assess a variety of aspects of RNA processing and regulation in the important human pathogen *Aspergillus fumigatus*. In addition, she has assessed the host Adenosine-to-Inosine RNA editing (a form of RNA modification) response to several human pathogenic fungi, including

Candida albicans and *Rhizopus oryzae*. Her main work can be framed in several ways, but centers around the double-stranded RNA molecule, both in its processing by enzymes like Dicer as part of RNA interference, and the ability of Adenosine deaminase acting on RNA (ADAR) proteins to deaminate adenosine to inosine as an RNA editing reaction. Both of these areas are at the cutting-edge of RNA research, particularly regarding fungal pathogens, and Ms. Pan's projects definitely fit this mold. By thorough investigations of these reactions in the context of pathogenic fungi, she has made intriguing observations that will result in several upcoming papers. One of these is already available as a preprint on *bioRxiv* and a second is focused on the host RNA editing and alternative splicing response to fungal pathogens that we believe has wide implications for understanding *Rhizopus* infections.

Ms. Pan has worked on several additional collaborations during her studies. She contributed to a study published several years ago in *RNA* further characterizing the RNA interference machinery of *A. fumigatus*, and she co-wrote a review article on the future of RNA modification research in infectious diseases published last year in *ACS Infectious Diseases*. Finally, she has contributed several analyses to a manuscript in preparation from the laboratory focused on the host extracellular RNA landscape in response to fungal infections. This is a long-standing project in the laboratory and will be submitted soon. These examples highlight the breadth of her work and show that Ms. Pan has been an important contributor and team player in her work performed in my laboratory over the past four years. I expect that her thesis will be submitted before the end of the year, as I can confirm many sections are already drafted.

Ms. Pan has been involved in numerous other training activities in her time at the Leibniz-HKI. She is a member of the Jena School for Microbial Communication, an affiliated graduate school. She is a regular attendee of the Jena RNA Club that my group co-organizes, and she also presented her work in this capacity at an annual symposium last year. My group joins a journal club organized together with the much larger department of Prof. Axel Brakhage, where Ms. Pan has presented numerous times, with great improvements over the years. In my own lab meeting, trainees are expected to present roughly once every two months, alongside a bimonthly in-depth review of a manuscript to practice critical assessment of the literature. Ms. Pan is a regular contributor, and never afraid to ask questions for clarification. Over the years, she has improved greatly as a scientific communicator as well. She now gives clear scientific talks and appreciates the advantages of telling a story with her research. Her spoken and written English is very good, as demonstrated by her willingness to contribute to the writing of drafts of her research manuscripts.

In sum, I have had the pleasure of watching Ms. Pan evolve from an inexperienced young scientist into an inquisitive, independent researcher over the past four years, and I look forward to seeing where she goes next in her career.

If you have additional questions, please do not hesitate to contact me by email (matthew.blango@leibniz-hki.de) or by phone (+49 03641 5321072).

Best Wishes,



Dr. Matthew G. Blango